Ziqi Wei

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EDUCATION

University of Illinois Chicago **Expected December 2025**
Master of Science in Civic Analytics GPA: 3.77

Beijing University of Posts and Telecommunications **Jun 2024**
Bachelor of Engineering in Information Security GPA: 3.3

SKILLS

- ***Programming & Tools*:** Python, R, VUE, Git&Github, ArcGIS Pro, SQL, Excel, BI tools
- ***AI/Machine Learning & Methods*:** Random Forest, XGBoost, SHAP, BERT, LSTM, K-Means, DBSCAN

RESEARCH EXPERIENCE

AI Integration Strategy Consulting for City Government **Aug 2025 – Dec 2025**

- Advised the City Administrator and leadership team of Warrenville on Artificial Intelligence adoption strategies through bi-weekly presentations and feedback sessions, ensuring alignment with municipal goals and operational needs.
- Conducted in-depth research on AI applications in comparable U.S. cities (e.g., Park Forest case study) and evaluated specific AI solutions (e.g., Citibot chatbot, Ordinal AI knowledge engine) for technical feasibility, cost-effectiveness, and potential integration challenges.
- Assessed departmental readiness and identified high-impact AI use cases across city operations (including Public Works, Administration, Finance), analyzing potential improvements in efficiency and public service delivery.
- Developed and prioritized actionable recommendations for AI implementation based on ROI, technical feasibility, and departmental needs, culminating in a strategic report and final presentation to guide city leadership's decision-making.

Text Sentiment Analysis with LSTM and BERT **May 2025**

- Developed and compared LSTM and fine-tuned BERT models to classify user reviews into positive, neutral, and negative sentiments.
- Conducted extensive text preprocessing including tokenization, stopword removal, and vectorization using TensorFlow and HuggingFace Transformers.
- Achieved over **88%** accuracy on the test set; conducted error analysis to address misclassifications and improve the model's robustness and interpretability.

Restaurant Review Rating Prediction **April 2025**

- Implemented supervised learning pipelines using Random Forest and XGBoost to predict restaurant review star ratings (1-5 scale) based on review text and associated categorical features, utilizing a dataset of **42K** Yelp reviews.
- Performed extensive text preprocessing including tokenization and stop word removal; engineered features from text (e.g., TF-IDF representation) and integrated categorical data to build robust predictive models.
- Executed systematic hyperparameter tuning to optimize model performance (evaluated using RMSE and R^2) and mitigate overfitting, achieved over **70%** prediction accuracy on the test set.
- Leveraged SHAP values to interpret model predictions at the review level, analyze the impact of specific words and features on predicted ratings, and enhance model transparency.

P2P Loan Default Prediction and Risk Analysis **March 2025**

- Built predictive models using Decision Trees to classify default risk for P2P loans, focusing on evaluating model discrimination ability.
- Performed extensive Exploratory Data Analysis to identify key risk indicators (e.g., loan grade, interest rate) and rigorously addressed data leakage issues, selected **67** relevant pre-loan predictors.
- Compared various Decision Tree configurations (including pruning, Gini/information gain, class balancing strategies) to optimize model performance and generalization.
- Identified the optimally pruned Decision Tree as the best model, achieving a Test AUC of over **65%**, demonstrated moderate predictive capability for default risk.